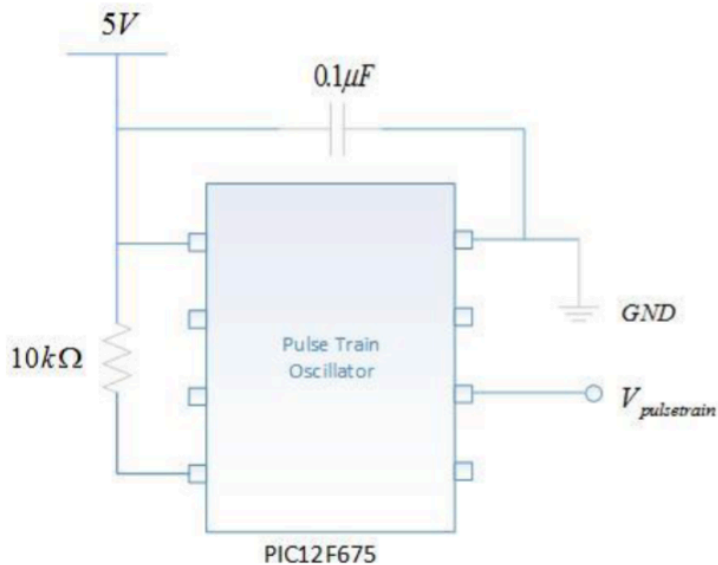


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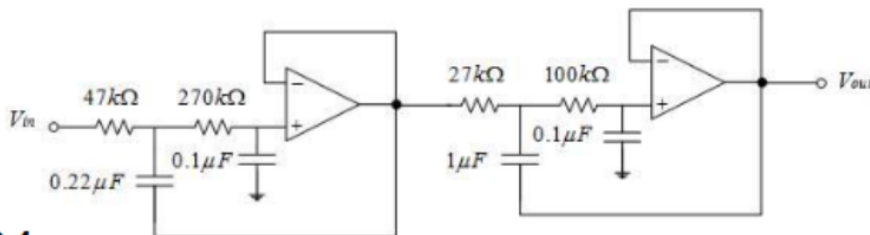
Design, build, and test a low pass filter to produce a 10 Hz sinusoidal signal from a 10 Hz pulse train.

We will make an analog and digital filter.

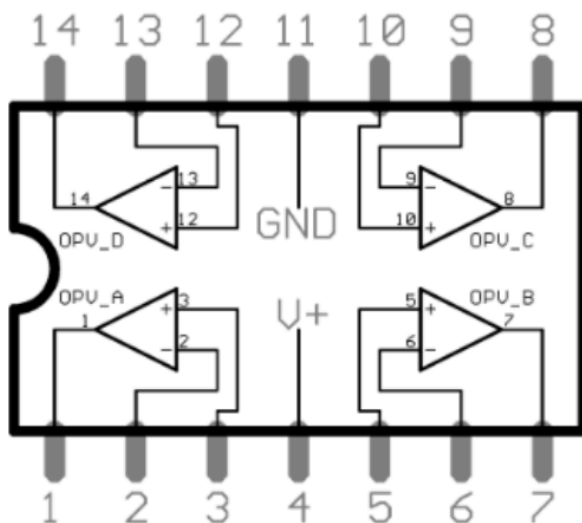
We will make a pulse train generator, then build a circuit for analog filtering, and then use the modified comb filter from homework 9b to filter in an Arduino.



We will build a 2 stage sallen key circuit.



Quad Op Amp: LMC6484



The pulse train has a fundamental frequency, but also higher harmonics. We are trying to get rid of those harmonics so that we can get a pure sinusoidal signal from the fundamental frequency.

Then we will use a digital filter.

Comb filter:

$$H(z) = b(1 - z^{-1})$$

And the modified comb filter to actually use

$$H(z) = \frac{1}{2}b_0(1 - e^{-j\frac{2\pi}{M}}z^{-1})$$

This is designed to eliminate harmonics at those harmonic frequencies.

We should get something close to a pure sine.